An Analysis of K-means clustering

K-means clustering

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

K-means clustering -- Part 1

a r g m i n S ■ ∑ i = 1 k 1 | S i | ∑ x , y ∈ S i ■ x − y ■ 2 {\displaystyle \mathop {\operatorname {arg\,min} } _{\mathbf {S} }\sum _{i=1}^{k}\,{\frac {1}{|S_{i}|}}\,\sum _{\mathbf {x} ,\mathbf {y} \in S_{i}}\left\|\mathbf {x} -\mathbf {y} \right\|^{2}}

K-means clustering -- Part 2

Modern astronomical surveys like APOGEE and the Gaia catalog collect spectroscopic and photometric data for millions of stars, making manual classification impractical at scale. K-means clustering has been applied to this data to automatically identify distinct stellar populations based on properties like luminosity, temperature, color, and chemical abundance. One major application is identifying stellar populations through chemical tagging. Stars that form together in the same cluster share similar chemical compositions, so clustering algorithms applied to abundance data can recover these groupings even after the stars have spread throughout the galaxy. Studies using k-means on APOGEE infrared spectra measured abundances of up to 13 elements and successfully distinguished known star clusters from surrounding field stars. This has also revealed populations with unusual abundance patterns that would be difficult to catch through manual inspection. K-means has also been combined with dimensionality reduction techniques to classify large numbers of unlabeled survey objects by separating stars, galaxies, and quasars based on their observed light properties. Labels from confirmed examples are then propagated to nearby cluster members. This approach has achieved accuracy comparable to fully supervised methods while requiring far fewer labeled examples, which makes it well suited for the scale of modern sky surveys.

K-means clustering -- Part 3

Recent developments Recent advancements in the application of k-means clustering have explored the integration of k-means clustering with deep learning methods, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to enhance the performance of various tasks in computer vision, natural language processing, and other domains.

K-means clustering -- Part 4

n is the number of d-dimensional vectors (to be clustered) k the number of clusters i the number of iterations needed until convergence. On data that does have a clustering structure, the number of iterations until convergence is often small, and results only improve slightly after the first dozen iterations. Lloyd's algorithm is therefore often considered to be of "linear" complexity in practice, although it is in the worst case superpolynomial when performed until convergence.

K-means clustering -- Part 5

Elbow method (clustering): This method involves plotting the explained variation as a function of the number of clusters, and picking the elbow of the curve as the number of clusters to use. However, the notion of an "elbow" is not well-defined and this is known to be unreliable. Silhouette (clustering): Silhouette analysis measures the quality of clustering and provides an insight into the separation distance between the resulting clusters. A higher silhouette score indicates that the object is well matched to its own cluster and poorly matched to neighboring clusters. Gap statistic: The Gap Statistic compares the total within intra-cluster variation for different values of k with their expected values under null reference distribution of the data. The optimal k is the value that yields the largest gap statistic. Davies–Bouldin index: The Davies-Bouldin index is a measure of the how much separation there is between clusters. Lower values of the Davies-Bouldin index indicate a model with better separation. Calinski-Harabasz index: This Index evaluates clusters based on their compactness and separation. The index is calculated using the ratio of between-cluster variance to within-cluster variance, with higher values indicate better-defined clusters. Rand index: It calculates the proportion of agreement between the two clusters, considering both the pairs of elements that are correctly assigned to the same or different clusters. Higher values indicate greater similarity and better clustering quality. To provide a more accurate measure, the Adjusted Rand Index (ARI), introduced by Hubert and Arabie in 1985, corrects the Rand Index by adjusting for the expected similarity of all pairings due to chance.

K-means clustering -- Part 6

Euclidean distance is used as a metric and variance is used as a measure of cluster scatter. The number of clusters k is an input parameter: an inappropriate choice of k may yield poor results. That is why, when performing k-means, it is important to run diagnostic checks for determining the number of clusters in the data set. Convergence to a local minimum may produce counterintuitive ("wrong") results (see example in Fig.). A key limitation of k-means is its cluster model. The concept is based on spherical clusters that are separable so that the mean converges towards the cluster center. The clusters are expected to be of similar size, so that the assignment to the nearest cluster center is the correct assignment. When for example applying k-means with a value of k = 3 {\displaystyle k=3} onto the well-known Iris flower data set, the result often fails to separate the three Iris species contained in the data set. With k = 2 {\displaystyle k=2} , the two visible clusters (one containing two species) will be discovered, whereas with k = 3 {\displaystyle k=3} one of the two clusters will be split into two even parts. In fact, k = 2 {\displaystyle k=2} is more appropriate for this data set, despite the data set's containing 3 classes. As with any other clustering algorithm, the k-means result makes assumptions that the data satisfy certain criteria. It works well on some data sets, and fails on others. The result of k-means can be seen as the Voronoi cells of the cluster means. Since data is split halfway between cluster means, this can lead to suboptimal splits as can be seen in the "mouse" example. The Gaussian models used by the expectation–maximization algorithm (arguably a generalization of k-means) are more flexible by having both variances and covariances. The EM result is thus able to accommodate clusters of variable size much better than k-means as well as correlated clusters (not in this example). In counterpart, EM requires the optimization of a larger number of free parameters and poses some methodological issues due to vanishing clusters or badly-conditioned covariance matrices. k-means is closely related to nonparametric Bayesian modeling.

K-means clustering -- Part 7

Initialization methods Commonly used initialization methods are Forgy and Random Partition. The Forgy method randomly chooses k observations from the dataset and uses these as the initial means. The Random Partition method first randomly assigns a cluster to each observation and then proceeds to the update step, thus computing the initial mean to be the centroid of the cluster's randomly assigned points. The Forgy method tends to spread the initial means out, while Random Partition places all of them close to the center of the data set. According to Hamerly et al., the Random Partition method is generally preferable for algorithms such as the k-harmonic means and fuzzy k-means. For expectation maximization and standard k-means algorithms, the Forgy method of initialization is preferable. A comprehensive study by Celebi et al., however, found that popular initialization methods such as Forgy, Random Partition, and Maximin often perform poorly, whereas Bradley and Fayyad's approach performs "consistently" in "the best group" and k-means++ performs "generally well".